Does the claw evolve?

LifeEngine · headless batch `claw-evolution` · 8 configurations × 10 runs · 150,000 ticks · 18 October 2026

> **In short:** the claw arises by mutation in every run, within about 1,000 ticks. It spreads only when burrowing works: from no claws, it took over in 7 of 10 runs when claws cost food (1 of 10 without burrowing, p = 0.02), and in 10 of 10 when claws are free (2 of 10 without, p = 0.0007). So the claw evolves *because* burrowing protects its carriers, not as free padding. With these settings, though, the spread rarely ends in collapse: all prey died after an evolved take-over in 2 of 20 runs.

Question and design

Earlier batches on this branch seeded 10 clawed prey at the start, so they couldn't show whether the claw would evolve and spread on its own. This batch starts with and without clawed prey, and adds controls that separate the claw's use for burrowing from its other effects.

Factor	Levels	What it tests
Starting prey	**seeded:** 10 prey + 10 clawed prey **evolved:** 20 prey, no claws	whether claws arise and spread by mutation alone
Claw cost	**free** (claw cells don't count toward the food to reproduce) or **cost** (they do)	whether free claws spread as padding rather than for burrowing
Burrowing	**burrow** (works) or **noburrow** (`minBurrowFertility` 11, above the maximum soil fertility of 10)	whether claws spread because burrowing helps; with no burrowing they can still evolve but do nothing

- **10 runs per configuration,** each with its own placement seed (1–10) and soil seed (42, 7, 123, 2024, 99, 11, 314, 555, 808, 1234), 384 × 193 grid, soil density 0 and patch resolution 20.
- **Predator lifespan multiplier 2,** following Tarique's report: at 2, prey survive the early predator wave without claws, so the claw has time to evolve. Everything else is at this branch's defaults: 10 predators and 39 plants, hides of 100 ticks costing the soil 10, no ageing while hidden, stay hidden near predators (range 6), only burrow when a predator is close (range 8), soil regrowth 0.05%.
- **Claws are counted per organism** (every prey carrying a claw). In the evolved starts every clawed prey has an evolved claw. In the seeded starts, evolved and seeded claws are separated by lineage, which is an estimate.
- **A take-over** follows the group's definition: at least 50 clawed prey making up at least half of all prey. **All prey died after a take-over** is the group's evolutionary-suicide outcome.

Results

Configuration	Claw evolved	First evolved claw (median tick)	Claws took over	Prey with a claw, late (median)	Lowest soil (median, % of start)	All prey died after take-over	World lasted
evolved, cost, burrow	**10/10**	**850**	**7/10**	**17%**	**86%**	**0**	**10/10**
evolved, cost, noburrow	10/10	750	1/10	1%	100%	0	10/10
evolved, free, burrow	**10/10**	**1,050**	**10/10**	**94%**	**73%**	**2**	**8/10**
evolved, free, noburrow	10/10	900	2/10	2%	100%	0	10/10
seeded, cost, burrow	10/10	1,450	9/10	38% seeded + 1% evolved	75%	1	9/10
seeded, cost, noburrow	10/10	850	2/10	about 0%	100%	0	10/10
seeded, free, burrow	10/10	2,050	10/10	90% seeded + 2% evolved	74%	2	8/10
seeded, free, noburrow	10/10	1,350	8/10	about 0%	100%	1	8/10

"Late" is the last 40% of each run. In the seeded starts, half of all prey are clawed at tick 0, so the take-over rule can trigger during the first prey boom; the late share is the better measure there.

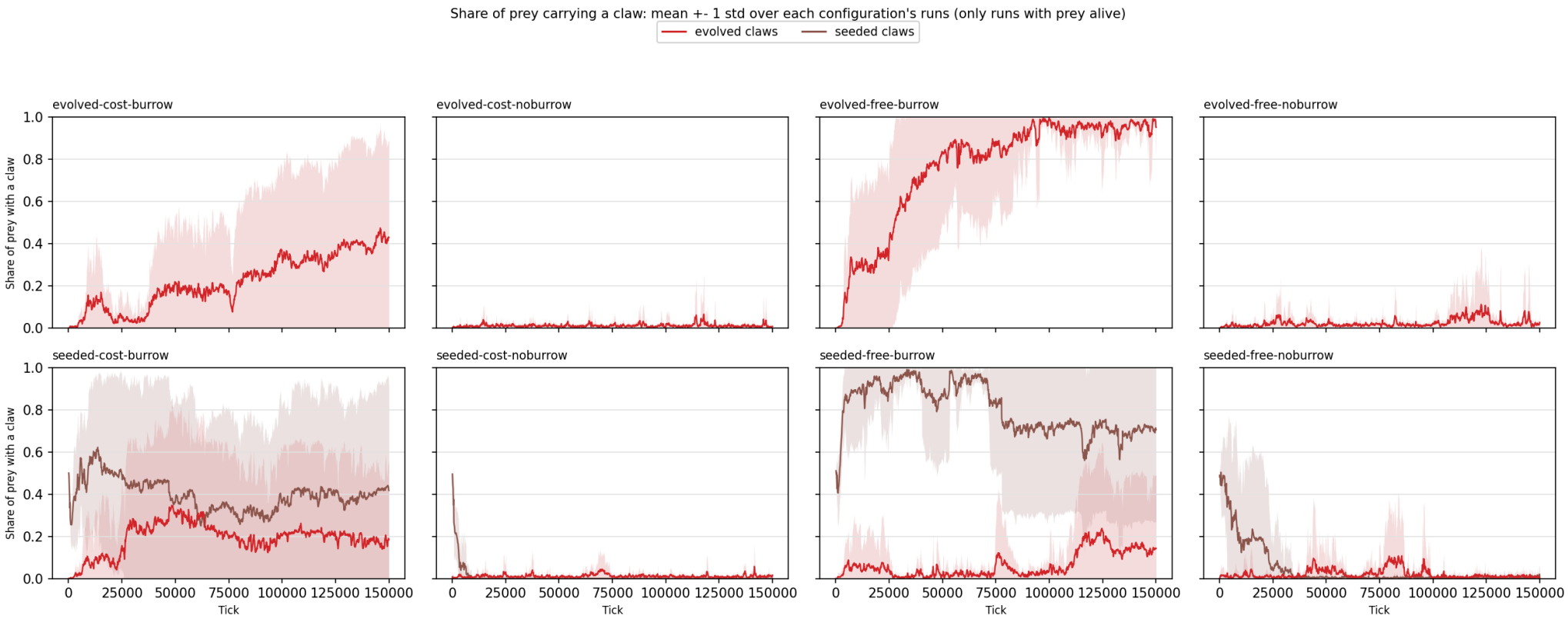


Figure 1. Share of prey carrying a claw over time, mean ± 1 std over each configuration's 10 runs. Red: evolved claws (arose by mutation). Brown: claws descended from the seeded clawed prey. Top row: no clawed prey at the start. Columns: claws cost food or are free, burrowing works or doesn't.

What it shows

1. **The claw arises by mutation in every run (80 of 80),** within a median 750–2,050 ticks. This matches Nik's soil study (median about 1,200 ticks).
2. **The claw spreads because burrowing helps.** From no claws, it took over in 7 of 10 runs with burrowing against 1 of 10 without when claws cost food (Fisher's exact test, p = 0.02), and 10 of 10 against 2 of 10 when claws are free (p = 0.0007). Without burrowing, claws stay on 1–2% of prey.
3. **Free claws don't spread among prey on their own.** Even seeded free claws fade to about 0% of prey by about 30,000 ticks when burrowing doesn't work (Figure 1, bottom right). In earlier batches predators and plants kept free claws; prey don't.
4. **Free claws make burrowing claws spread much further:** to 94% of prey against 17% when claws cost food. The difference in take-overs (10/10 against 7/10) isn't significant (p = 0.21), but the late share is very different. Claws costing food remain the cleaner test, as in the group's reports.
5. **Burrowing wears the soil down** (to a median 73% of its starting fertility with free evolved claws, 86% when they cost food) and plants fall to a median low of 134, against 450 without burrowing.
6. **Collapses after a take-over are rare here.** All prey died after evolved claws took over in 2 of 10 runs (free claws, at ticks 64,100 and 102,400) and never when claws cost food.

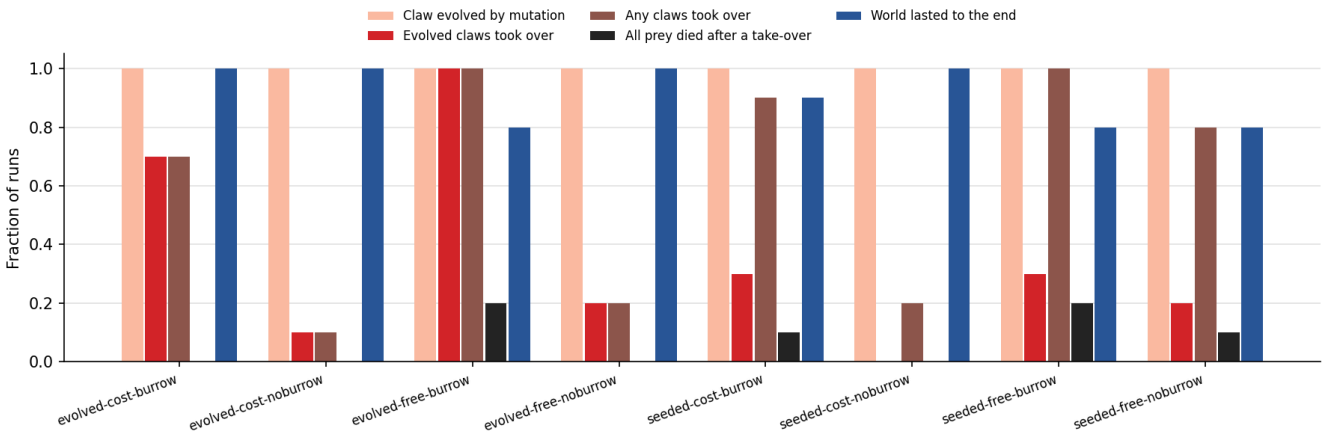


Figure 2. Per configuration, the fraction of runs where a claw evolved by mutation, evolved claws took over, any claws took over, all prey died after a take-over, and the world lasted 150,000 ticks.

How it compares with the group's reports

- **Agrees:** claws evolve quickly by mutation (Nik), and spread when burrowing works but not when it is disabled (Tarique: 19 of 20 take-overs with harmless burrowing; here 7 of 10 from no claws when claws cost food).
- **Fewer collapses:** Tarique found all prey dying after a take-over in 21 of 40 runs; here 0 of 10 with claws costing food. The main difference is *Only burrow when predator is close*: off in Tarique's runs, on here (this branch's default). With it on, prey run from distant predators instead of burrowing, so the soil is damaged less (lowest soil 86% here against about 70% in Tarique's runs).
- **New here:** the free-claw factor shows that free claws don't spread among prey without burrowing, and the seeded starts show evolved claws arising alongside seeded ones (1–2% of prey late in the run).

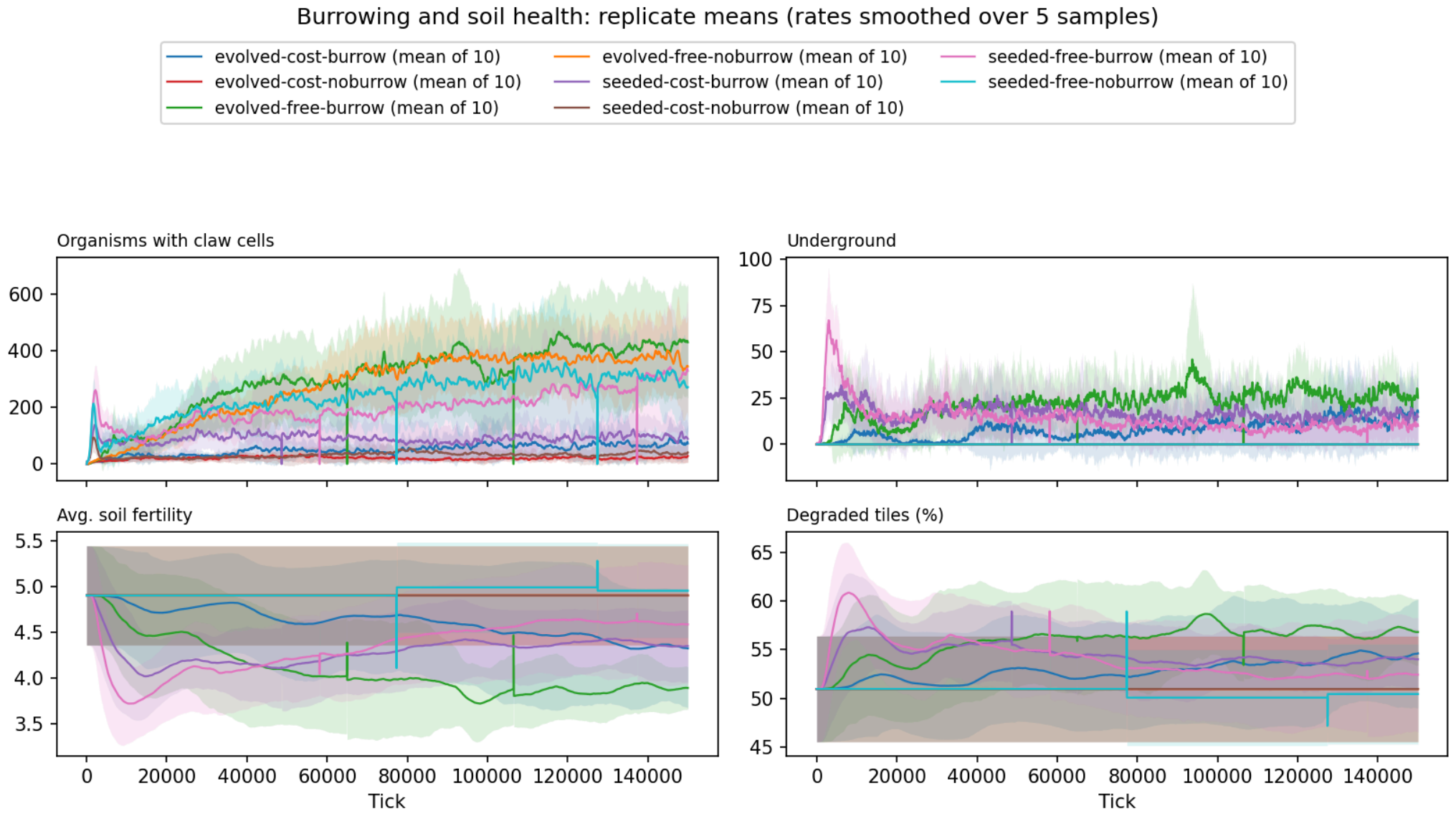


Figure 3. Burrowing and soil health, replicate means (± 1 std). Organisms with claw cells include predators and plants, which keep free claws even when burrowing is off. Soil bands are wide because each run has its own soil layout.

Caveats

- 10 runs per configuration; differences of 2 or 3 runs could be chance. Only the burrow against no-burrow comparisons are clearly significant.
- In the seeded starts, the split between evolved and seeded claws is estimated from species lineages.
- One soil density and patch size (0 / 20), though each run has its own soil seed.
- Not measured: whether clawed prey still out-reproduce clawless prey while a population declines (the part of the evolutionary-suicide definition Tarique also marked as untested).

Next steps

1. **Rerun the evolved starts with *Only burrow when predator is close* off**, to test whether evolved claws lead to collapse as often on this branch as in Tarique's runs.
2. **Add a harmless-burrowing control** (`hideFertilityCost` 0), to separate the claw's spread from its soil damage, as in the group's reports.
3. **Measure fitness of clawed against clawless prey during declines**, to complete the evolutionary-suicide definition.

Reproducing

`node headless/batch.js headless/experiments/claw_evolution.json` (about 35 minutes on this machine), then `python analysis/claw_evolution.py Observations/claw-evolution` for the claw measures and `python analysis/plot_metrics.py Observations/claw-evolution` for the standard plots. Per-run results are in `Observations/claw-evolution/claw_evolution/runs.csv`.